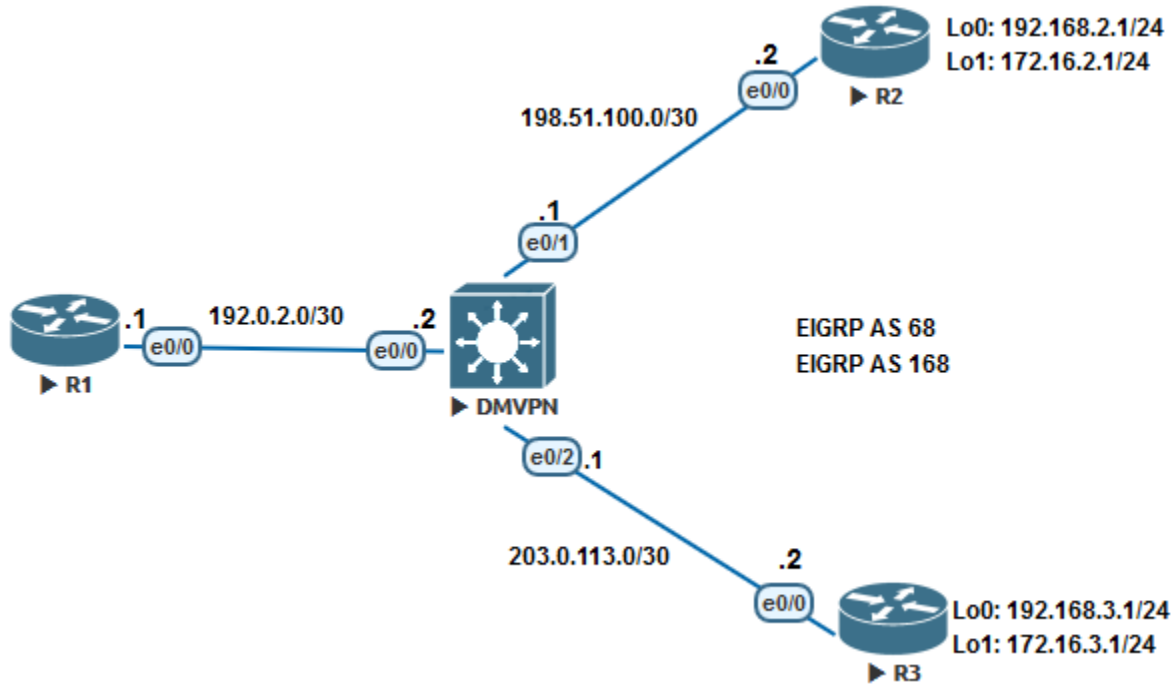


Lab - Configure Secure DMVPN Tunnels



Addressing Table

Device	Interface	IPv4 Address
R1	E0/0	192.0.2.1/24
	Tunnel 1	100.100.100.1/29
R2	E0/0	198.51.100.2/24
	Loopback 0	192.168.1.1/24
	Loopback 1	172.16.1.1/24
	Tunnel 1	100.100.100.2/29
R3	E0/0	203.0.113.2/24
	Loopback 0	192.168.3.1/24
	Loopback 1	172.16.3.1/24
	Tunnel 1	100.100.100.3/29

Objectives

Part 1: Build the Network and Verify DMVPN Phase 3 Operation

Part 2: Secure DMVPN Phase 3 Tunnels

Background / Scenario

In previous labs, you have configured DMVPN Phase 1 and Phase 3 networks, including configuration of DMVPN Phase 3 with IPv6. However, in those labs, IPsec was not used to encrypt and protect data travelling on the tunnels. IPsec functionality is essential to DMVPN implementation. In this lab, you will work with the DMVPN Phase 3 implementation from the Implement a DMVPN Phase 3 Spoke-to-Spoke Topology lab. You will start with a working configuration and then apply IPsec to the spoke-to-hub and spoke-to-spoke tunnels. Finally, you will verify the operation of the secured tunnels.

Part 1: Build the Network and Verify DMVPN Phase 3 Operation

In Part 1, you will set up the network topology and configure basic settings if the network is not already configured. This lab uses the same topology and final configurations from the **Implement a DMVPN Phase 3 Spoke-to-Spoke Topology** lab.

Step 1: Configure initial settings for each router and the Layer 3 switch.

Console into each device, enter global configuration mode, and apply the initial settings for the lab if the devices are not already configured.

Step 2: Verify connectivity in the network.

- a. From R1, **ping** the loopback interfaces of R2 and R3. All pings should be successful. This verifies that full connectivity exists in the underlay, or transport, network.

```
R1# ping 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

R1# ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Step 3: Verify DMVPN Phase 3 operation.

- a. Return to R2. Initiate a **traceroute** to the simulated LAN interface on R3. The path will pass through R1 as it does in a DMVPN Phase 1 network.

Note: The first trace may fail if the DMVPN switch CAM table is empty.

```
R2# traceroute 172.16.3.1
Type escape sequence to abort.
Tracing the route to 172.16.3.1
VRF info: (vrf in name/id, vrf out name/id)
 1 100.100.100.1 1 msec 1 msec 1 msec
 2 100.100.100.3 1 msec * 2 msec
```

- b. Issue the **traceroute** command again. You will now see that R1 has enabled direct spoke-to-spoke communication between R2 and R3. This tunnel will expire and close dynamically. The tunnel reopens after data for the spoke router is sent again.

```
R2# traceroute 172.16.3.1
Type escape sequence to abort.
Tracing the route to 172.16.3.1
VRF info: (vrf in name/id, vrf out name/id)
 1 100.100.100.3 1 msec * 1 msec
```

Part 2: Secure DMVPN Phase 3 Tunnels

Now that the tunnels have been configured and DMVPN connectivity has been verified, the tunnels can be secured with IPsec.

Step 1: Create the IKE policy.

Create an IKE policy that defines the hash algorithm, encryption type, key exchange method, Diffie-Hellman group, and the authentication method.

```
R1(config)# crypto isakmp policy 99
R1(config-isakmp)# hash sha384
R1(config-isakmp)# encryption aes 256
R1(config-isakmp)# group 14
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# exit
```

Step 2: Configure the ISAKMP key.

Configure the pre-shared key and peer address. Use 0.0.0.0 to match multiple peer addresses. Use a key of **DMVPN@key#**.

```
R1(config)# crypto isakmp key DMVPN@key# address 0.0.0.0
```

Step 3: Create and configure the IPsec transform set.

Configure the IPsec transform set. Use **DMVPN_TRANS** as the transform set name. Specify **esp-aes** with a 256-bit key as the encryption transform and **esp-sha384-hmac** as the authentication transform. Configure the transform set to use IPsec **transport** mode for the tunnels.

```
R1(config)# crypto ipsec transform-set DMVPN_TRANS esp-aes 256 esp-sha384-hmac
R1(cfg-crypto-trans)# mode transport
R1(cfg-crypto-trans)# exit
```

Step 4: Create the IPsec profile.

Create an IPsec profile with the name **DMVPN_PROFILE**. Associate the **DMVPN_TRANS** transform set with the profile.

```
R1(config)# crypto ipsec profile DMVPN_PROFILE
R1(ipsec-profile)# set transform-set DMVPN_TRANS
R1(ipsec-profile)# exit
```

Step 5: Apply the IPsec profile to the tunnel interface.

Finally, apply the IPsec profile to the tunnel interface. After you apply the profile, you will see that IPsec is now active and you will lose adjacency with R2 and R3 until their respective ends of the tunnel are configured.

```
R1(config)# interface tunnel 1
R1(config-if)# tunnel protection ipsec profile DMVPN_PROFILE
R1(config-if)# exit
```

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```
*Mar 30 07:39:32.398: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config)#
*Mar 30 07:39:32.963: %IOSXE-3-PLATFORM: R0/0: cpp_cp: QFP:0.0 Thread:001
TS:00000000594132950499 %IPSEC-3-RECVD_PKT_NOT_IPSEC: Rec'd packet not an IPSEC
packet, dest_addr= 192.0.2.1, src_addr= 192.168.2.1, prot= 47
*Mar 30 07:39:43.664: %DUAL-5-NBRCHANGE: EIGRP-IPv4 68: Neighbor 100.100.100.2
(Tunnel1) is down: holding time expired
*Mar 30 07:39:44.235: %DUAL-5-NBRCHANGE: EIGRP-IPv4 68: Neighbor 100.100.100.3
(Tunnel1) is down: holding time expired
R1(config)#
```

Step 6: Configure R2 and R3 with IPsec.

Repeat this configuration on the R2 and R3 routers.

Step 7: Verify DMVPN Phase 3 operation.

- a. As was done previously, test the operation of the spoke-to-spoke DMVPN. Return to R2. Initiate a **traceroute** to the simulated LAN interface on R3. The path will pass through R1 as it does in a DMVPN Phase 1 network.

```
R2# traceroute 172.16.3.1
Type escape sequence to abort.
Tracing the route to 172.16.3.1
VRF info: (vrf in name/id, vrf out name/id)
 1 100.100.100.1 1 msec 1 msec 1 msec
 2 100.100.100.3 1 msec * 2 msec
```

- b. Issue the **traceroute** command again. You will now see that R1 has enabled direct spoke-to-spoke communication between R2 and R3. This tunnel will expire and close dynamically. The tunnel reopens after data for the spoke router is sent again.

```
R2# traceroute 172.16.3.1
Type escape sequence to abort.
Tracing the route to 172.16.3.1
VRF info: (vrf in name/id, vrf out name/id)
 1 100.100.100.3 1 msec * 1 msec
```

Step 8: Verify IPsec configuration.

Note: Shut down a tunnel interface to clear its IPsec socket if you wish to explore the outputs before and after spoke-to-spoke tunnel establishment.

- a. To show information about the IPsec profiles that are configured on a device, issue the **show crypto ipsec profile** command. Note that the profile that was previously configured is shown along with a default profile.

```
R2# show crypto ipsec profile
IPSEC profile DMVPN_PROFILE
    Security association lifetime: 4608000 kilobytes/3600 seconds
    Responder-Only (Y/N): N
    PFS (Y/N): N
    Mixed-mode : Disabled
    Transform sets={
        DMVPN_TRANS: { esp-256-aes esp-sha384-hmac } ,
    }
```

```
IPSEC profile default
  Security association lifetime: 4608000 kilobytes/3600 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): N
  Mixed-mode : Disabled
  Transform sets={
    default: { esp-aes esp-sha-hmac } ,
  }
```

- b. It is very important to verify that tunnel traffic will be encrypted. On R1, issue the **show dmvpn detail** command. As the hub router, R1 should see the spoke peers. The first part of the output shows the tunnel interface status and the peer table. Both peers should be shown with their transport and overlay interface addresses, as you have seen previously.

The Crypto Session Details portion of the output should contain information about the status of the encrypted tunnels. Both of the spoke routers should appear in this output also. Note that the transform set that you configured is also displayed in the Crypto Session output.

R1# **show dmvpn detail**

<output omitted>

```
Interface Tunnel1 is up/up, Addr. is 100.100.100.1, VRF ""
  Tunnel Src./Dest. addr: 192.0.2.1/Multipoint, Tunnel VRF ""
  Protocol/Transport: "multi-GRE/IP", Protect "DMVPN_PROFILE"
  Interface State Control: Disabled
  nhrp event-publisher : Disabled
Type:Hub, Total NBMA Peers (v4/v6): 2
```

#	Ent	Peer NBMA Addr	Peer Tunnel Add	State	UpDn	Tm	Attrb	Target Network
1		192.168.2.1	100.100.100.2	UP	00:04:25		D	100.100.100.2/32
1		192.168.3.1	100.100.100.3	UP	00:04:59		D	100.100.100.3/32

Crypto Session Details:

Interface: Tunnel1

```
Session: [0x7F6E17B867D0]
  Session ID: 0
  IKEv1 SA: local 192.0.2.1/500 remote 192.168.2.1/500 Active
    Capabilities:(none) connid:1001 lifetime:23:59:19
  Session ID: 0
  IKEv1 SA: local 192.0.2.1/500 remote 192.168.2.1/500 Active
    Capabilities:(none) connid:1002 lifetime:23:59:28
Crypto Session Status: UP-ACTIVE
fvrf: (none), Phase1_id: 192.168.2.1
IPSEC FLOW: permit 47 host 192.0.2.1 host 192.168.2.1
  Active SAs: 4, origin: crypto map
  Inbound: #pkts dec'ed 17 drop 0 life (KB/Sec) 4607998/3568
  Outbound: #pkts enc'ed 16 drop 0 life (KB/Sec) 4607999/3568
  Outbound SPI : 0xD2E76488, transform : esp-256-aes esp-sha384-hmac
```

Socket State: Open

Interface: Tunnel1

Session: [0x7F6E17B86950]

Session ID: 0

IKEv1 SA: local 192.0.2.1/500 remote 192.168.3.1/500 Active

Capabilities:(none) connid:1004 lifetime:23:59:48

Session ID: 0

IKEv1 SA: local 192.0.2.1/500 remote 192.168.3.1/500 Active

Capabilities:(none) connid:1003 lifetime:23:59:40

Crypto Session Status: UP-ACTIVE

fvrfr: (none), Phase1_id: 192.168.3.1

IPSEC FLOW: permit 47 host 192.0.2.1 host 192.168.3.1

Active SAs: 6, origin: crypto map

Inbound: #pkts dec'ed 11 drop 0 life (KB/Sec) 4607999/3588

Outbound: #pkts enc'ed 10 drop 0 life (KB/Sec) 4607999/3588

Outbound SPI : 0xCB3D3313, transform : esp-256-aes esp-sha384-hmac

Socket State: Open

Pending DMVPN Sessions:

- c. Issue the **show crypto ipsec sa** command on R2 to display the security associations (sa) that have been made by R2. This output is for the spoke-to-hub tunnel between R1 and R2 prior to the establishment of the spoke-to-spoke tunnel. This command provides additional details regarding the IPsec status of the tunnel, encrypted and decrypted packet statistics, and other details regarding characteristics of the encrypted tunnel.

R2# **show crypto ipsec sa**

interface: Tunnel1

Crypto map tag: Tunnel1-head-0, local addr 192.168.2.1

protected vrf: (none)

local ident (addr/mask/prot/port): (192.168.2.1/255.255.255.255/47/0)

remote ident (addr/mask/prot/port): (192.0.2.1/255.255.255.255/47/0)

current_peer 192.0.2.1 port 500

PERMIT, flags={origin_is_acl,}

#pkts encaps: 125, #pkts encrypt: 125, #pkts digest: 125

#pkts decaps: 126, #pkts decrypt: 126, #pkts verify: 126

#pkts compressed: 0, #pkts decompressed: 0

#pkts not compressed: 0, #pkts compr. failed: 0

#pkts not decompressed: 0, #pkts decompress failed: 0

#send errors 0, #recv errors 0

local crypto endpt.: 192.168.2.1, remote crypto endpt.: 192.0.2.1

plaintext mtu 1458, path mtu 1514, ip mtu 1514, ip mtu idb Loopback0

current outbound spi: 0x97C1D18A(2546061706)

PFS (Y/N): N, DH group: none

inbound esp sas:

spi: 0xD2E76488(3538379912)

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```
transform: esp-256-aes esp-sha384-hmac ,
in use settings ={Transport, }
conn id: 2003, flow_id: ESG:3, sibling_flags FFFFFFFF80000008, crypto map:
Tunnell-head-0
sa timing: remaining key lifetime (k/sec): (4607984/3047)
IV size: 16 bytes
replay detection support: Y
Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcp sas:

outbound esp sas:
spi: 0x97C1D18A(2546061706)
transform: esp-256-aes esp-sha384-hmac ,
in use settings ={Transport, }
conn id: 2004, flow_id: ESG:4, sibling_flags FFFFFFFF80000008, crypto map:
Tunnell-head-0
sa timing: remaining key lifetime (k/sec): (4607990/3047)
IV size: 16 bytes
replay detection support: Y
Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcp sas:
```

The output below is for the same command after the spoke-to-spoke tunnel is open. Entries exist for both the tunnel to R1 and the spoke-to-spoke tunnel between R2 and R3.

R2# **show crypto ipsec sa**

```
interface: Tunnell
Crypto map tag: Tunnell-head-0, local addr 192.168.2.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.2.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (192.168.3.1/255.255.255.255/47/0)
current_peer 192.168.3.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
  #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 192.168.2.1, remote crypto endpt.: 192.168.3.1
plaintext mtu 1458, path mtu 1514, ip mtu 1514, ip mtu idb Loopback0
current outbound spi: 0x658E8CF5(1703841013)
```

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```
PFS (Y/N): N, DH group: none

inbound esp sas:
  spi: 0xFA8FC9F2(4203727346)
    transform: esp-256-aes esp-sha384-hmac ,
    in use settings ={Transport, }
    conn id: 2005, flow_id: ESG:5, sibling_flags FFFFFFFF80000008, crypto map:
Tunnell-head-0
    sa timing: remaining key lifetime (k/sec): (4608000/3316)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)
  spi: 0x59C41A42(1506024002)
    transform: esp-256-aes esp-sha384-hmac ,
    in use settings ={Transport, }
    conn id: 2007, flow_id: ESG:7, sibling_flags FFFFFFFF80004008, crypto map:
Tunnell-head-0
    sa timing: remaining key lifetime (k/sec): (4608000/3326)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0x60CC6F77(1624010615)
    transform: esp-256-aes esp-sha384-hmac ,
    in use settings ={Transport, }
    conn id: 2006, flow_id: ESG:6, sibling_flags FFFFFFFF80000008, crypto map:
Tunnell-head-0
    sa timing: remaining key lifetime (k/sec): (4608000/3316)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)
  spi: 0x658E8CF5(1703841013)
    transform: esp-256-aes esp-sha384-hmac ,
    in use settings ={Transport, }
    conn id: 2008, flow_id: ESG:8, sibling_flags FFFFFFFF80004008, crypto map:
Tunnell-head-0
    sa timing: remaining key lifetime (k/sec): (4608000/3326)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcp sas:
```

```
protected vrf: (none)
local  ident (addr/mask/prot/port): (192.168.2.1/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (192.0.2.1/255.255.255.255/47/0)
current_peer 192.0.2.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 67, #pkts encrypt: 67, #pkts digest: 67
#pkts decaps: 67, #pkts decrypt: 67, #pkts verify: 67
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

local crypto endpt.: 192.168.2.1, remote crypto endpt.: 192.0.2.1
plaintext mtu 1458, path mtu 1514, ip mtu 1514, ip mtu idb Loopback0
current outbound spi: 0x97C1D18A(2546061706)
PFS (Y/N): N, DH group: none

inbound esp sas:
  spi: 0xD2E76488(3538379912)
    transform: esp-256-aes esp-sha384-hmac ,
    in use settings ={Transport, }
    conn id: 2003, flow_id: ESG:3, sibling_flags FFFFFFFF80000008, crypto map:
Tunnell-head-0
    sa timing: remaining key lifetime (k/sec): (4607991/3305)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcg sas:

outbound esp sas:
  spi: 0x97C1D18A(2546061706)
    transform: esp-256-aes esp-sha384-hmac ,
    in use settings ={Transport, }
    conn id: 2004, flow_id: ESG:4, sibling_flags FFFFFFFF80000008, crypto map:
Tunnell-head-0
    sa timing: remaining key lifetime (k/sec): (4607995/3305)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcg sas:
```

- d. On R2 issue the **show crypto isakmp sa** command to view the Internet Security Association Management Protocol (ISAKMP) SAs between the peers. Before the formation of the spoke-to-spoke tunnel, SAs have been made between R2 and R3, but no further negotiations have occurred, as indicated by the MM_NO_STATE state of the two SAs between the routers.

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```
R2# show crypto isakmp sa
```

```
IPv4 Crypto ISAKMP SA
```

dst	src	state	conn-id	status
192.0.2.1	192.168.2.1	QM_IDLE	1001	ACTIVE
192.168.2.1	192.0.2.1	QM_IDLE	1002	ACTIVE
192.168.3.1	192.168.2.1	MM_NO_STATE	1004	ACTIVE (deleted)
192.168.2.1	192.168.3.1	MM_NO_STATE	1003	ACTIVE (deleted)

```
IPv6 Crypto ISAKMP SA
```

After traffic has established the spoke-to-spoke tunnel, the SAs all show the QM_IDLE state. The SAs have been fully negotiated and are available for further ISAKMP quick mode exchanges.

Note: ISAKMP modes are outside the scope of this course.

```
R2# show crypto isakmp sa
```

```
IPv4 Crypto ISAKMP SA
```

dst	src	state	conn-id	status
192.0.2.1	192.168.2.1	QM_IDLE	1001	ACTIVE
192.168.2.1	192.0.2.1	QM_IDLE	1002	ACTIVE
192.168.3.1	192.168.2.1	QM_IDLE	1004	ACTIVE
192.168.2.1	192.168.3.1	QM_IDLE	1003	ACTIVE

```
IPv6 Crypto ISAKMP SA
```

- e. You have successfully configured and verified IPsec on DMVPN Phase 3 tunnels.